Shared factorial errors in a Diophantine equation for Con(ZFC)

A direct quartic with 240 unrestricted integer unknowns

Prepared for Vladimir Reshetnikov and the ProveIt project
Mathematical development and computational assistance: OpenAI ChatGPT

25 October 2026

Abstract

We give a completely specified degree-four polynomial in 240 unrestricted integer unknowns whose lack of integer zeros is provably equivalent in ZFC to Con(ZFC). The final equation has 177 quadratic residuals, 21482 collected monomials, and coefficient height 6144. Every residual is printed in this report.

The main mathematical change shares one norm certificate between eleven modular digit remainders and a signed factorial approximation error. A nearest-integer factorial lemma recovers the intended factorial after binomial extraction, making two former positivity certificates unnecessary. Centered digits also allow an initially signed odd radix, whose positivity is recovered later. The original beta-spacing formula is retained. These changes give a 244-variable direct core; exact integral elimination and reversible residual-row operations produce the stated final equation.

The route uses the numerical proof calculus C222 and its identified ZF/ZFC refutation calibration, without instantiating K215 or Jones's universal polynomial. We distinguish ordinary unbounded proofs from finite algebra checks. No global simplicity minimum, consistency proof, or newly completed end-to-end proof-assistant formalization is claimed.

Contents

1 An explicit equation and the scope of the result **3**

2 The retained arithmetic interface **4**

3 Pell orientation before arithmetic signs **5**

4 A power lemma valid at signed bases **6**

5 One norm also certifies the factorial error **6**

6 Recovering the radix and the first exact digit **7**

6.1 Excluding the two unit radices 7

6.2 A binomial coefficient at a possibly negative radix 7

7 The nearest-integer factorial fixed point **8**

8 The remaining digits and the calibrated refutation **9**

9 Constructive completeness **10**

10 From the source to the delivered quartics **11**

11 What has and has not been established **12**

A The complete residual list **13**

B Coordinate names and artifact interpretation **26**

1 An explicit equation and the scope of the result

The principal deliverable is the following single polynomial equation over unrestricted integers:

$$P(x_0, \dots, x_{239}) := \sum_{i=0}^{176} Q_i(x_0, \dots, x_{239})^2 = 0. \tag{1}$$

The appendix gives every coefficient of every quadratic polynomial Q_i. The file `equation240/equation.json` is the same finite definition in a sparse, machine-readable format; the accompanying collected coefficient file gives P directly. There are no unspecified universal polynomials, variable-exponent operations, or machine simulations in this definition. The powers appearing in the proof below describe the meanings of ordinary polynomial registers.

Theorem 1.1. *For the fixed integer polynomial in (1), ZFC proves*

$$(\forall \mathbf{X} \in \mathbb{Z}^{240} \quad P(\mathbf{X}) \neq 0) \quad \Longleftrightarrow \quad \mathrm{Con}(\mathrm{ZFC}).$$

More precisely, ZFC proves that an integer zero exists if and only if a finite refutation exists in the fixed numerical presentation of ZFC used by the retained ProveIt calibration.

The equation has 240 variables, 177 squared quadratic residuals, total degree four, 21482 nonzero collected monomials, and coefficient height 6144. The height is the largest absolute coefficient after all like monomials have been collected. All variables range over $\mathbb{Z}$, including coordinates whose eventual arithmetic interpretations are nonnegative.

The comparison is with the incoming common-remainder construction at the pinned repository snapshot, rather than with an older, larger ancestor. The exact endpoints supplied here make the tradeoffs visible:

Equation	Variables	Residuals	Monomials	Height
Common-remainder parent	252	182	5340	128446
New arithmetic core	244	181	5346	128446
Primary dimension endpoint	240	177	21482	6144
Balanced companion	241	178	5351	15107
Lower-height companion	241	178	9544	4512

All five entries have degree four. The primary endpoint gives the smallest dimension among the new supplied forms, at a substantial cost in support: its collected polynomial has about four times as many terms as the 252-variable parent. The balanced companion keeps almost the parent's support while removing eleven variables and reducing coefficient height. These are distinct exact equations with the same existence theorem.

This is an improvement of the direct numerical proof-calculus branch. It does not establish a global minimum. Other branches of the repository already contain quartics in far fewer variables, including a 37-integer form, with different coefficient and encoding costs [6]. There is no established scalar measure called *simplest* that orders dimension, degree, coefficient size, support, and proof complexity. Here the declared objective is an explicit quartic with small coefficients, a direct refutation interpretation, and a complete reproducible description.

The new arithmetic step is to use the existing common norm to certify the factorial error as well as all eleven digit remainders. This removes the two initial factorial sign certificates. A signed-radix proof then permits the radix certificate to be only an affine oddness equation. These changes are compatible with the inherited deterministic Pell comparison center. The common normalized remainders, the center construction, and the underlying calculus are credited to their predecessors [2, 4, 3, 1]; the proof below identifies precisely which additional implications are needed.

2 The retained arithmetic interface

Write $t = 16 + d^2$, where d is an unrestricted integer. The principal arithmetic quantities are

$$F, \quad B, \quad V, \quad x, \quad y, \quad N, \quad q, \quad M, \qquad H = 1 + \sum_{\nu=1}^{9} a_\nu.$$

The a_ν are the nine actual CRT representatives. In the stored source, several are a coordinate plus one; it is their actual values, rather than their shifted coordinates, that will become nonnegative.

The retained exact-power requests have the intended meanings

$$V = (F^2 + 1)^t, \qquad N = B^x, \qquad q = (BH)^x, \qquad M = q^x, \tag{2}$$

together with the powers used in the binomial numerators and denominators. There are eighteen requests, sharing fourteen independent principal Pell pairs. At this point these displays are requested meanings, not assumptions. Their interpretation is recovered in a specified order below.

The beta spacing is left literally unchanged:

$$y = V(V - 1). \tag{3}$$

The endpoint equations include

$$x = F + 1 + (y + 1)u_0, \qquad x = 1 + \big(1 + (t + 1)y\big)u_t. \tag{4}$$

Initially u_0, u_t, x, F, B, q may have either sign.

The eleven normalized remainder registers fall into four groups:

Denominator	Meaning	Number
q^t	Factorial digit and encoded index	2
q^{t+1}	Four syntax representatives and derived label	5
q^{y+1}	Two pending-multiset representatives	2
q^x	Two beta quotients	2

The existing ratio equations are

$$q^t v_t = M, \qquad q^{t+1} v_{t+1} = M, \qquad q^{y+1} v_y = M. \tag{5}$$

Each displayed power in these equations is a retained output register. The quotient for the last group is the constant one.

For clarity, put

$$G(a, b, c) = a^2 + b^2 + c(c + 1).$$

This form is nonnegative at every integer triple and represents every nonnegative integer. The latter is the familiar consequence of the three-square theorem applied to $4n + 1$: exactly one square is odd, and the two even squares can be divided by four. This is the same guard form already used in the parent [2].

The altered arithmetic core replaces the three separate initial scalar sign certificates by

$$q = 2z + 1, \tag{6}$$

$$Bv = M, \tag{7}$$

$$MF = vV + S, \tag{8}$$

$$M^2 = 1 + 4\left(S^2 + \sum_{i=1}^{11} R_i^2\right) + G(a, b, c). \tag{9}$$

The new integers v, S normalize the factorial error at the same denominator M as the other remainders. No inequality, division, factorial, or variable power has been added to the literal polynomial system. Equations (6)–(9) are polynomial rows of degree at most two in their actual registers.

3 Pell orientation before arithmetic signs

For a signed integer center A, define the Pell polynomials by

$$X_0(A) = 1, \quad Y_0(A) = 0, \qquad X_{k+1} = AX_k + (A^2 - 1)Y_k, \quad Y_{k+1} = X_k + AY_k.$$

For negative indices use $X_{-k} = X_k$, $Y_{-k} = -Y_k$. The retained frame bounds every actual base, requested index, and proposed power output in absolute value by an integer W, with

$$W \geq 258, \qquad |A| > W^W, \qquad |A - 1| > 4W^3. \tag{10}$$

These are consequences of square-based frame equations. Their proof does not assume signs for F, B, q, x, or the power outputs.

The deterministic comparison center is retained from the supplied predecessor proof [4]. To distinguish its auxiliary pair from the arithmetic quantity H, write it as

$$H_{\mathrm{p}}^2 - (A^2 - 1)J_{\mathrm{p}}^2 = 1, \qquad J_{\mathrm{p}} = C^2(2e + 1),$$

where (U, C) is the principal anchor pair with requested index h. The squared cap says

$$C^2 = h^2 + 1 + \sum_{\text{other pairs}} Y_i^2 + G(a', b', c').$$

It gives $C \neq 0$, $|h| < |C|$, and strict domination of all other principal ordinates. The comparison center is defined by

$$f = (1 - A)H_{\mathrm{p}}, \qquad D = A + H_{\mathrm{p}}f = 1 - (A - 1)(A^2 - 1)J_{\mathrm{p}}^2.$$

Thus $|D| > 1$, $D \equiv A \pmod{H_{\mathrm{p}}}$, and $D \equiv 1 \pmod{C}$. The comparison ordinate residue is $R = h + Ck$.

Here is the part of its proof that matters for compatibility. Classify the anchor and common Pell ordinates at $|A|$ by positive indices i, l. For $i \geq 2$, the squared divisibility condition implies $|C| \mid l$ and $l > i$. The abscissa comparison gives the comparison index $j \equiv \pm i \pmod{2l}$; the new ordinate residue gives $j \equiv \pm h \pmod{|C|}$. Consequently $i \equiv \pm h \pmod{|C|}$. The growth bound

$$|C| = Y_i(|A|) \geq i + 2|A| - 2$$

and $|h| < W$ imply $i + |h| < |C|$, so $i = |h| < W$. All other classified indices are smaller by the cap.

The exceptional case $i = 1$ has $|C| = 1$. The cap then forces $h = 0$ and every other ordinate to vanish. Reducing the retained orientation pool modulo $A - 1$ gives a nonnegative small error sum. Its anchor ordinate contributes one, so it is strictly positive, while (10) makes it strictly smaller than $|A - 1|$. Its asserted divisibility is impossible. This exceptional branch therefore supplies no zero.

All fourteen principal pairs retain both-coordinate orientation: the squared abscissa errors and squared ordinate-index errors remain in the pool. After classification their small residues have errors at most two and strictly less than $2W$, respectively. The resulting nonnegative residue sum is less than $4 \cdot 14(W^2 + 1) < 4W^3 < |A - 1|$. Divisibility forces every error to vanish. Thus each principal pair is exactly $(X_k(A), Y_k(A))$ at its signed requested index.

This use of all fourteen pairs is deliberate. A separate predecessor reduction keeps fewer abscissa tests, but its hypotheses cannot simply be assumed in the signed bootstrap below. The present construction does not make that deletion. The complete retained center and frame proofs are included in the proof closure.

4 A power lemma valid at signed bases

Lemma 4.1. *Assume (10). Let $|n|, |k|, |m| < W$, with $|n| \geq 2$, and set*

$$T = 2An - n^2 - 1.$$

If the pair is canonically oriented at k and its power row asserts

$$X_k(A) + (n - A)Y_k(A) \equiv m \pmod{T},$$

then $k \geq 0$ and $m = n^k$. For $n = \epsilon \in \{1, -1\}$, the same row instead gives $m = \epsilon^k$ for every integer k, including negative k.

Proof. For $|n| \geq 2$,

$$|T| > 4W^W - (W - 1)^2 - 1 > W^W + W.$$

Modulo T, one has $(n - A)^2 = A^2 - 1$ and $n(2A - n) = 1$. The Pell identities therefore give n^k at nonnegative indices and its multiplicative inverse at negative indices. If $k \geq 0$, the congruence has

$$|m - n^k| < W + W^W < |T|,$$

so it is equality. If $k = -j < 0$, multiplication by n^j gives $mn^j \equiv 1 \pmod{T}$. But

$$|mn^j - 1| \leq (W - 1)^W + 1 < |T|,$$

forcing the impossible integer equality $mn^j = 1$.

For a unit base, $T = 2A\epsilon - 2$ and $2A \equiv 2\epsilon \pmod{T}$. The sequence $f_k = X_k(A) + (\epsilon - A)Y_k(A)$ has initial values $f_0 = 1, f_1 = \epsilon$ and satisfies $f_{k+1} = 2Af_k - f_{k-1}$. This recurrence runs backwards as well as forwards. Hence $f_k \equiv \epsilon^k \pmod{T}$ for every signed integer k. Finally $|T| > 2W^W - 2 > W$, whereas $|m - \epsilon^k| \leq W$, giving the stated equality. □

The base -1 case is essential: it cannot be replaced by a lemma that presupposes a nonnegative exponent. The literal request for $M = q^x$ has the row

$$-M + X_x(A) - AY_x(A) + qY_x(A) - T_q\eta = 0.$$

Both M and q, and the actual base BH of the self-radix request, are included in the frame. Thus the lemma applies to the unshifted arithmetic values claimed here.

5 One norm also certifies the factorial error

The shared-radius argument works before the denominator signs are known. Suppose nonzero integers L, v_i, M satisfy $Lv_i = M$ and

$$v_iU = MQ + R_i.$$

Then $v_i \mid R_i$; setting $r_i = R_i/v_i$ gives an integer and recovers $U = LQ + r_i$. Equation (9) implies

$$|R_i| < |M|/2, \qquad |r_i| < |L|/2.$$

If a prescribed signed remainder r_i^0 is congruent to U modulo L and has $|r_i^0| < |L|/2$, uniqueness follows: the multiple $r_i - r_i^0$ of L has absolute value less than $|L|$. Positivity of either L or r_i^0 is unnecessary.

The same mechanism yields the new factorial estimate. First, (9) gives $M \neq 0$. Equations (7)–(8) then imply

$$B \neq 0, \quad v \neq 0, \quad S = v(BF - V).$$

With $b = |B|$ and $s = BF - V$, this gives

$$s \in \mathbb{Z}, \qquad |s| < b/2. \tag{11}$$

The canonical pair at the already positive index t gives $V = (F^2 + 1)^t \geq 1$, using the unit case if $F = 0$.

Lemma 5.1. *Under these conditions, $b \geq 2$, $b > |F|$, and $V < b^2$. If $F = 0$, in fact $b \geq 3$. If $F \neq 0$, B and F have the same sign.*

Proof. When $F = 0$, $V = 1$ and $|BF - V| = 1 < b/2$, so $b > 2$. Suppose $f = |F| \geq 1$. Opposite signs would give $|BF - V| = bf + V > b/2$, a contradiction. Therefore $BF = bf$. If $b \leq f$, (11) would give

$$V < b(f + 1/2) \leq f^2 + f/2,$$

contradicting $V \geq (f^2 + 1)^2 > f^2 + f/2$. Thus $b \geq f + 1 \geq 2$, and

$$V < b(f + 1/2) \leq b(b - 1/2) < b^2.$$

□

This estimate supplies the coefficient margin formerly obtained from two separate nonnegativity graphs. It has not yet identified B as a binomial coefficient or F as a factorial.

6 Recovering the radix and the first exact digit

6.1 Excluding the two unit radices

Equation (6) makes q odd, hence nonzero. If $q = 1$ or $q = -1$, Lemma 4.1 applied to the request $M = q^x$ gives $|M| = 1$, with no assumption on x. But $Bv = M$ and $|B| \geq 2$ make this impossible. Consequently

$$Q := |q| \geq 3, \qquad |q + 1| \geq 2.$$

Every actual representative a_ν is a requested exponent at base $q + 1$. The same lemma now gives $a_\nu \geq 0$, and therefore $H \geq 1$. The derived label h is also an exponent at that base, so $h \geq 0$; the anchor conclusion excludes $h = 0$. The request at base q gives $x \geq 0$ and $M = q^x$.

The retained ratio $q^{t+1} v_{t+1} = M$ is now an equality of nonzero integer powers. Its absolute values imply $x \geq t + 1$. Since $|BH| \geq b \geq 2$, the self-radix request gives

$$q = (BH)^x, \qquad Q \geq b^x \geq b^{t+1}. \tag{12}$$

Neither the radix nor B has been assumed positive in this argument. The beta value (3) is nonnegative, but its size is not needed to reach (12).

6.2 A binomial coefficient at a possibly negative radix

Put $n = F^2 + t$ and $C_j = \binom{n}{j}$. The exact numerator request at base $q + 1$ gives $U = (q + 1)^n$. Dividing its normalized digit row by its nonzero ratio gives

$$U = q^t(B + q\eta) + r, \qquad |r| < Q^t/2. \tag{13}$$

For $F \neq 0$, every coefficient through position t satisfies

$$C_j \leq (F^2+1)^t = V < b^2 \qquad (0 \leq j \leq t). \tag{14}$$

For completeness, the general inequality $\binom{m+t}{j} \leq (m+1)^t$, for $m \geq 1$ and $j \leq t$, follows by induction on t. Pascal's identity handles $j \leq t$ in the next row using $2 \leq m+1$; the new edge is bounded by $\binom{m+t+1}{t+1} = \prod_{i=1}^{t+1}(1+m/i) \leq (m+1)^{t+1}$. If $F = 0$, use the separate bound $C_j \leq 2^t$. Using $V = 1$ to bound those coefficients would be incorrect.

For either sign of q, the binomial theorem gives the signed low polynomial

$$r^0 = \sum_{j=0}^{t-1} C_j q^j, \qquad |r^0| < \frac{C_{\max}}{Q-1} Q^t.$$

When $F \neq 0$, (12) gives

$$\frac{C_{\max}}{Q-1} < \frac{2}{b^{t-1}} < \frac{1}{32}.$$

When $F = 0$, $b \geq 3$, and

$$\frac{C_{\max}}{Q-1} \leq \frac{2^t}{3^{t+1}-1} < (2/3)^{t+1} < \frac{1}{32} \qquad (t \geq 16).$$

Thus $|r^0| < Q^t/32$. It is congruent to U modulo the possibly negative integer q^t. The signed remainder uniqueness argument gives $r = r^0$.

Subtract the low polynomial from (13), divide by q^t, and reduce modulo q. The result is

$$B \equiv C_t \pmod{q}.$$

In the nonzero-F case, $|B - C_t| \leq b + C_t < b + b^2 < Q$. In the zero-$F$ case, $C_t = 1$ and $b + 1 < Q$. Therefore in both cases

$$B = \binom{F^2+t}{t} > 0. \tag{15}$$

The case $F = 0$ would give $B = 1$, contradicting $b \geq 3$. Hence $F \neq 0$; the sign conclusion of Lemma 5.1 now gives $F > 0$. Finally $q = (BH)^x$ with $B > 0, H \geq 1, x \geq t+1$ gives $q > 0$. The negative-radix branch has served its purpose and has been excluded by the arithmetic itself. All remaining exact-power requests, including $N = B^x$, now have their usual positive-base meanings.

7 The nearest-integer factorial fixed point

The new bound is symmetric around zero, so the old ceiling lemma alone does not justify the construction. The required strengthened statement has a short direct proof.

Lemma 7.1. *For integers $t \geq 16$, $F \geq 1$, put*

$$R_t(F) = \frac{(F^2+1)^t}{\binom{F^2+t}{t}}.$$

Then

$$|F - R_t(F)| < \tfrac{1}{2} \quad \Longleftrightarrow \quad F = t!.$$

At the solution one also has

$$0 < F - R_t(F) < \tfrac{1}{32}.$$

Proof. Write

$$R_t(F) = t! \prod_{i=2}^{t} \left(1 - \frac{i-1}{F^2+i}\right).$$

Every factor increases with F. Consequently

$$R_t(F) \geq R_t(1) = \frac{2^t}{t+1} > t^2 + \tfrac{1}{2} \qquad (t \geq 16).$$

The last inequality holds at 16 and persists by elementary induction. The assumed nearest-integer bound therefore gives $F > t^2$. Using $\prod_i (1-e_i) \geq 1 - \sum_i e_i$ for $0 \leq e_i \leq 1$, we obtain

$$R_t(F) > t! \left(1 - \frac{t(t-1)}{2F^2}\right) > \frac{3}{4} t!.$$

Thus $F > R_t(F) - 1/2 > t!/2$. Applying the same estimate a second time, now at this much larger lower bound, gives

$$0 < t! - R_t(F) < \frac{t!\, t(t-1)}{2F^2} < \frac{2t(t-1)}{t!} < \frac{1}{32}.$$

The last bound holds at 16, and its left-hand expression decreases by a factor $1/(t-1)$ when t increases by one. It follows that

$$|F - t!| \leq |F - R_t(F)| + |R_t(F) - t!| < \frac{17}{32} < 1.$$

Both F and $t!$ are integers, so $F = t!$.

Conversely, set $F = t!$. The same product estimate directly gives

$$0 < F - R_t(F) < \frac{t(t-1)}{2t!} < \frac{1}{32},$$

which implies the nearest-integer bound. □

Equations (11) and (15) provide exactly the hypothesis of Lemma 7.1. Hence

$$F = t!, \qquad B = \prod_{i=1}^{t} \left(1 + \frac{F^2}{i}\right), \qquad 0 < BF - V < B/32. \tag{16}$$

The old nonnegative factorial slacks can now be reconstructed as $s = BF - V$ and $r = B - s - 1$. Both are nonnegative integers. Also $q \geq B^{t+1} > B$, so the old output slack $q - B - 1$ is nonnegative. Fresh G-witnesses can therefore fill each deleted scalar certificate. Their signs are conclusions of the proof.

8 The remaining digits and the calibrated refutation

The beta spacing was not enlarged or otherwise modified. From $V = (F^2+1)^t$ one has $F^2 \mid V - 1$, and therefore $F^2 \mid y$. The standard cofinal CRT proof for $y = V(V-1)$ applies literally. In particular $y > F + 2$. In the first endpoint equation, $u_0 \leq -1$ would imply $x \leq F - y < 0$, contradicting $x \geq t + 1$. Thus $u_0 \geq 0$ and $x \geq F + 1$. The second endpoint, together with $x > 1$, gives $u_t \geq 1$ and

$$x \geq 2 + (t+1)y. \tag{17}$$

All four range denominators in the table are now positive divisors of $M = q^x$.

For reference, the retained strict amplification estimates are stronger than mere denominator divisibility. Set $\mathcal{R} = 2 + t + x + y + F$. Equation (17) and $y > F + 2$ give $\mathcal{R} < 2x$, while $x \geq F + 1 \geq 16! + 1 > 2^{32}$. The inherited local polynomial ledger uses exponent 2^{23}, and

$$2^{23} \log_2(2x) < x \qquad (x \geq 2^{32}).$$

Hence its local bound is less than $\mathcal{R}^{2^{23}} < (2x)^{2^{23}} < 2^x$. This is exactly the amplification margin used in the parent prime-power extraction, with precision $2x$ and loss less than x.

After the factorial interpretation, the encoded-index congruence is $B \mid C_0 + F^2 + 1$. Since $B > F^2 + 1$, the nonnegative representative C_0 cannot vanish; consequently $H \geq 2$. For any ordinary representative $a \geq 0$ and relevant range position $k \leq x$, use

$$\binom{a}{j} \leq \max(1, a)^k \leq H^x \qquad (0 \leq j \leq k).$$

The maximum is necessary when $a = j = 0$. For the derived label, the retained degree ledger gives $h \leq H^{2^{22}}$, and the factorial and endpoint bounds give $2^{22}(t+1) < x$. Its required coefficients have the same upper bound H^x.

As $q = B^x H^x$, every true low remainder for the other ten digits is nonnegative and satisfies

$$r_i^0 < \frac{H^x}{q-1} q^{k_i} < \frac{2}{B^x} q^{k_i} < q^{k_i}/32.$$

The shared norm and normalized equations therefore recover these ordinary low remainders exactly. Their quotient congruences are precisely the parent's N-divisibility range tests, with $N = B^x \mid q$. The inherited prime-power argument recovers the bounded local parameters. The constructor equations and Horn-flow identities are unchanged, so they decode a finite refutation in the numerical C222 calculus [2, 1].

This is a statement about the actual nonnegative semantic representatives. A stored representative-minus-one coordinate may equal -1; no pointwise claim that all old natural coordinates have become nonnegative is needed or made.

The retained logical interfaces are named

`KEndpoint.c222_iff_zf_refutation`

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

Their ordinary calibration identifies existence of the decoded refutation with existence of a finite ZFC refutation in the fixed numeration. The retained source proofs explain the underlying syntax, rules, and calibration. This completes the soundness direction at the arithmetic source level.

9 Constructive completeness

Conversely, begin with a finite calibrated refutation. The inherited cofinal construction chooses an arbitrarily large horizon $t = 16 + d^2$, puts $F = t!$, $V = (F^2 + 1)^t$, $y = V(V - 1)$, and supplies a sufficiently large beta-CRT code x. The two genuine beta quotients are nonnegative and less than x, since their denominators are at least two. The strict range tests therefore admit the same canonical witnesses as the parent.

There are two parity conditions on the amplified representatives. The factors $1 + F^2/i$ are all odd, because $F^2/i = F(F/i)$ is even. Hence B, $N = B^x$, and the precision modulus N^2 are odd. Changing a representative by N^2 preserves its CRT residues and reverses its parity.

Choose the dominant representative a in $[2N^2, 4N^2)$, and choose b with opposite parity. The retained literal constructors give

$$h \geq 2a^2, \qquad h \equiv 1 + a + b \pmod 2.$$

Thus h is even and remains a dominant semantic anchor. Choose the parity of a third independent representative so that the sum of all nine representatives is even. Each other representative can remain in $(0, 2N^2]$. These choices preserve the original dominance and precision estimates and make H odd. Now $q = (BH)^x$ is odd, and

$$z = (q - 1)/2$$

is the required integer witness for (6). There is no need to prescribe the parity of x. This is weaker than the inherited odd-square construction [3], whose parity argument supplied an even x.

For completeness of the deterministic center, take a positive admissible A, $C = Y_h(A)$, and $U = X_h(A)$. Since h is even, C is even and U is odd. Choose $l = hC$. Pell composition gives

$$J_{\mathrm{p}} = Y_l(A) = C\, Y_C(U).$$

The polynomial expansion and $U^2 - 1 = (A^2 - 1)C^2$ give

$$\frac{Y_C(U)}{C} \equiv U^{C-1} \pmod C.$$

Thus J_{p}/C^2 is an odd integer, supplying e. Take $H_{\mathrm{p}} = X_l(A)$, $f = (1 - A)H_{\mathrm{p}}$, and $D = A + H_{\mathrm{p}} f$. The comparison pair $(X_h(D), Y_h(D))$ has the required abscissa congruence modulo H_{p} and ordinate congruence modulo C. All quotients are integers, even if D is negative. The retained cofinal frame and squared-cap construction supplies its other witnesses. The full orientation pool holds because canonical Pell coordinates have residues $1, h$ modulo $A - 1$.

Finally $B \mid q \mid M = q^x$. Set

$$v = M/B, \qquad S = v(BF - V).$$

Lemma 7.1 gives

$$0 < S/M = F - V/B < 1/32.$$

For each genuine digit remainder choose $R_i = (M/L_i)r_i^0$. The preceding bounds give $0 \leq R_i < M/32$. Consequently

$$4\left(S^2 + \sum_{i=1}^{11} R_i^2\right) < \frac{48}{1024} M^2 < M^2.$$

The integer $M^2 - 1 - 4(S^2 + \sum R_i^2)$ is nonnegative and therefore has a G-representation. This supplies the single new norm row. All other arithmetic, constructor, and Horn rows retain their canonical witnesses. Every finite refutation thus yields an integer zero of the new source.

10 From the source to the delivered quartics

The source-level modifications above are justified by the unbounded arithmetic proof. The subsequent reductions are exact finite algebra and require no further number-theoretic premise [5]. Their certificates distinguish three operations.

First, an affine integer coordinate shear or translation is a bijection of the integer lattice, with its stated integral inverse. Second, an elementary residual replacement

$$Q_i \longleftarrow Q_i + g(\mathbf{X})Q_j, \qquad i \neq j,$$

preserves the common zero set and has the explicit inverse with $-g$; the final coefficient reductions use integer row shears. Third, a residual $X_j - f(\widehat{\mathbf{X}}) = 0$ with unit coefficient allows integral elimination of X_j, with the unique restoration $X_j = f(\widehat{\mathbf{X}})$. Each actual substitution and degree bound is checked on its literal coefficient list.

The primary endpoint uses six coordinate shears, five elementary exposure actions, four unit-affine projections, and 409 final unimodular residual shears after the arithmetic core. These operations explain why dimension can fall while the final sum of squares becomes denser and its coefficients become smaller. An invertible row operation preserves simultaneous zeros; it need not preserve the numerical value of the sum of squares. The polynomial in (1) is the sum of squares of the final rows, not an assertion that all intermediate quartics are coefficientwise identical.

The supplied verifier replays the finite transformations, checks their inverses or unit pivots, translates coordinates by the stated convention, expands every squared residual, collects every coefficient, and recomputes the reported metrics. The accompanying receipts record the completed checks and rejection controls. The source snapshot and retained proof dependencies are pinned; the construction does not depend on a moving repository branch.

Over the integers, a sum of squares is zero exactly when every summand is zero. Combining this fact, the certified finite transformations, source soundness, source completeness, and the retained logical calibration gives

$$\mathrm{ZFC} \vdash \left[\exists \mathbf{X} \in \mathbb{Z}^{240}\ P(\mathbf{X}) = 0 \iff \exists p\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot)\right].$$

Negation yields Theorem 1.1.

11 What has and has not been established

The completed mathematical contribution is a shared certificate for the factorial error and digit errors, together with a signed-radix bootstrap that removes the need for the original three initial scalar sign graphs. The delicate points are explicit: both unit radices are excluded before assuming a nonnegative exponent; the first binomial digit is recovered in negative radix by absolute remainder uniqueness; the $F = 0$ coefficients are bounded separately; and the symmetric error condition is proved to force $F = t!$ by a nearest-integer theorem. The beta spacing and all fourteen principal orientations remain intact.

The result is an ordinary mathematical proof, composed with identified inherited arithmetic and logical proofs, and supported by complete finite algebraic artifacts. It is formalizable in ZFC. The new unbounded argument has not been emitted as a complete proof-assistant kernel term or as a literal object-language ZFC derivation. The finite checker authenticates the polynomial and its algebraic maps; it does not prove the universal Pell, CRT, or calibration theorems by numerical testing. Internal AI-assisted review is documented as such and is not external peer review.

No consistency assumption is used in the equivalence, and no decision of the equation's solvability or proof of Con(ZFC) is claimed. The improvement is substantial within its declared direct quartic architecture, while its global mathematical novelty and optimality remain separate questions. Its concrete output is the explicit family of equations, the proof of the new arithmetic reduction, and the reproducible coefficient and transformation data.

References

[1] V. Reshetnikov and ProveIt contributors, *ProveIt*, inspected commit `3ecd6a6cf53d47519228328567f45fb7ee9781d9`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9.

[2] ProveIt incoming contribution, *ConZFC Common Remainders Radix47*, 25 October 2026. Retained 252-integer direct equation and proof closure. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Common_Remainders_Radix47_2026-10-25.zip.

[3] ProveIt incoming contribution, *ConZFC Centered Residues*, 25 October 2026. Source of the odd-square radix predecessor and the inspected compact 75/626, 76/464, and 77/460 points. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Centered_Residues_2026-10-25.zip.

[4] ProveIt incoming contribution, *ConZFC Symmetric Remainders*, 25 October 2026. The deterministic-center proof is retained; its selected-nine orientation optimization is not used here. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Symmetric_Remainders_2026-10-25.zip.

[5] ProveIt incoming contribution, *ConZFC Residual Bases and Denominator Engineering*, 25 October 2026. Source of the compared affine-exposure and residual-basis methodology. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Residual_Bases_Denominator_Engineering_2026-10-25.zip.

[6] ProveIt, *Con(ZFC) in Diophantine form*, collection overview, and *Later arrivals and current size reductions*, active review, at the inspected commit. Counts are compared in their stated domains. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9/Computability/DiophantineConsistency.

A The complete residual list

The variables $x_0, \ldots, x_{239}$ range independently over $\mathbb{Z}$. The entire equation is $P = \sum_{j=0}^{176} Q_j^2 = 0$. The following list is generated from the literal JSON and independently parsed back into its exact integer coefficient data. Each monomial coefficient is literal. There are no additional quantified registers.

$$Q_0 = -x_3^2 + x_4$$

$$Q_1 = x_0 - x_0\,x_2 - x_2 + x_9$$

$$Q_2 = -x_2\,x_7 - x_3 + x_8\,x_9$$

$$Q_3 = -1 + x_1 - x_8\,x_9$$

$$Q_4 = 1 + x_4 - x_5\,x_{20} + x_{18}$$

$$Q_5 = -1 - x_{10}^2 - 2\,x_{10}\,x_{11} - x_{11}^2 + x_{21}$$

$$\begin{aligned} Q_6 = {} & 2 + x_1 - x_3\,x_{11} - x_4 + x_5\,x_{20} - x_8\,x_9 \\ & + x_{10} + x_{10}^2 - x_{10}\,x_{11} + x_{10}\,x_{12} - 8\,x_{10}\,x_{21} \\ & - 4\,x_{10}\,x_{25} - 12\,x_{11} - 5\,x_{11}^2 + 7\,x_{11}\,x_{12} - 16\,x_{11}\,x_{21} \\ & - x_{18} - 4\,x_{21} + 12\,x_{21}^2 + x_{22} + x_{25} + x_{26} + 2\,x_{27} \\ & - x_{40} - x_{48} + x_{78} + x_{83} \end{aligned}$$

$$\begin{aligned} Q_7 = {} & -1 - x_1 + x_8\,x_9 + 2\,x_{10} - x_{10}^2 - 4\,x_{10}\,x_{22} - 4\,x_{22} \\ & - 4\,x_{22}^2 + x_{23} \end{aligned}$$

$$
\begin{aligned}
Q_{8} = {} & -24\,x_{10}\,x_{23} + 24\,x_{10}\,x_{37} + 2\,x_{23} - 4\,x_{23}^{2} + x_{24} - 2\,x_{37} \\
& + 4\,x_{37}^{2} - x_{38}
\end{aligned}
$$

$$
\begin{aligned}
Q_{9} = {} & -1 - x_{1} + x_{8}\,x_{9} - x_{11} - x_{11}^{2} - 2\,x_{11}\,x_{12} - x_{12} \\
& - x_{12}^{2} + x_{25}
\end{aligned}
$$

$$
\begin{aligned}
Q_{10} = {} & -2 - x_{1} + x_{3}\,x_{11} + x_{8}\,x_{9} + 7\,x_{10} - x_{10}^{2} \\
& + x_{10}\,x_{11} + 3\,x_{10}\,x_{12} - 4\,x_{10}\,x_{25} + x_{11} + 2\,x_{11}^{2} \\
& + 3\,x_{11}\,x_{12} - x_{12} + 3\,x_{12}^{2} + 2\,x_{25} + x_{26} - x_{48} - x_{78} \\
& - x_{83}
\end{aligned}
$$

$$
Q_{11} = -1 + x_{10} - x_{10}^{2} - 2\,x_{10}\,x_{12} + x_{12} - x_{12}^{2} + x_{27}
$$

$$
\begin{aligned}
Q_{12} = {} & 1 + 2\,x_{0} + x_{1} - x_{3}\,x_{11} - x_{8}\,x_{9} + x_{10} + 4\,x_{10}^{2} \\
& - 13\,x_{10}\,x_{11} + x_{10}\,x_{12} - 16\,x_{10}\,x_{21} + 8\,x_{10}\,x_{65} + 4\,x_{11} \\
& + 4\,x_{11}^{2} - 9\,x_{11}\,x_{12} + 8\,x_{11}\,x_{25} - 4\,x_{11}\,x_{27} + 16\,x_{11}\,x_{65} \\
& + 6\,x_{12} + 7\,x_{12}^{2} - 2\,x_{12}\,x_{13} - 24\,x_{12}\,x_{21} - x_{13}^{2} - 13\,x_{21} \\
& + 8\,x_{21}^{2} - 8\,x_{21}\,x_{27} - 8\,x_{21}\,x_{65} + 6\,x_{25} - 4\,x_{25}^{2} \\
& - 8\,x_{25}\,x_{27} + 2\,x_{26} + 7\,x_{27} - 4\,x_{27}^{2} + x_{28} + x_{30} - x_{33} \\
& - 8\,x_{35} - 8\,x_{35}^{2} + 2\,x_{36} - x_{40} + x_{43} - 2\,x_{48} + 4\,x_{65} \\
& - 4\,x_{65}^{2} + x_{66} + 7\,x_{78} + x_{83} - 2\,x_{214}^{2}
\end{aligned}
$$

$$
Q_{13} = -4\,x_{26}^{2} - 8\,x_{26}\,x_{28} - 4\,x_{28}^{2} + x_{29}
$$

$$
\begin{aligned}
Q_{14} = {} & -2 - x_{1} + x_{3}\,x_{11} + x_{8}\,x_{9} - x_{10} - x_{10}^{2} \\
& + 4\,x_{10}\,x_{12} + 4\,x_{10}\,x_{25} + 8\,x_{10}\,x_{27} - 5\,x_{11} + 3\,x_{11}^{2} \\
& + 6\,x_{11}\,x_{12} + 8\,x_{11}\,x_{25} + 8\,x_{11}\,x_{27} - x_{12} + 3\,x_{12}^{2} \\
& - 4\,x_{21} + 9\,x_{25} - 4\,x_{25}^{2} - 8\,x_{25}\,x_{27} + x_{26} + 8\,x_{27} \\
& - 4\,x_{27}^{2} + x_{30} - x_{48} - x_{83}
\end{aligned}
$$

$$
\begin{aligned}
Q_{15} = {} & -1 - x_{1} + x_{3}\,x_{11} + x_{8}\,x_{9} - 7\,x_{10} + 7\,x_{10}^{2} \\
& + 7\,x_{10}\,x_{11} - 3\,x_{10}\,x_{12} + 8\,x_{10}\,x_{21} + 4\,x_{10}\,x_{25} \\
& - 8\,x_{10}\,x_{27} - 8\,x_{10}\,x_{30} + 2\,x_{11} - 15\,x_{11}^{2} + 11\,x_{11}\,x_{12} \\
& - 16\,x_{11}\,x_{21} + 8\,x_{11}\,x_{25} - 24\,x_{11}\,x_{27} - 16\,x_{11}\,x_{30} - x_{12} \\
& + 4\,x_{12}^{2} - 2\,x_{12}\,x_{13} + 8\,x_{12}\,x_{27} + 4\,x_{12}\,x_{30} - x_{13}^{2} \\
& + 2\,x_{21} - 4\,x_{21}^{2} - 16\,x_{21}\,x_{27} - 8\,x_{21}\,x_{30} + 7\,x_{25} - 4\,x_{25}^{2} \\
& - 8\,x_{25}\,x_{27} + x_{26} + 16\,x_{27} - 4\,x_{27}^{2} + 5\,x_{30} + x_{31} \\
& + x_{43} - x_{47} - x_{48} + x_{78} - x_{83}
\end{aligned}
$$

$$
\begin{aligned}
Q_{16} = {} & 2 + 2\,x_{0} + x_{1} - x_{3}\,x_{11} - x_{4} + x_{5}\,x_{20} \\
& - x_{8}\,x_{9} - 2\,x_{10} - x_{10}^{2} - 7\,x_{10}\,x_{11} - 5\,x_{10}\,x_{12} \\
& + 8\,x_{10}\,x_{27} + 4\,x_{10}\,x_{57} - 11\,x_{11} + 9\,x_{11}^{2} + x_{11}\,x_{12} \\
& - 8\,x_{11}\,x_{25} + 8\,x_{11}\,x_{27} - 16\,x_{12} - 2\,x_{12}^{2} - 8\,x_{12}\,x_{25} \\
& + 8\,x_{12}\,x_{57} - x_{18} - x_{21} - 12\,x_{25} - 16\,x_{25}^{2} - 8\,x_{25}\,x_{27} \\
& + x_{26} + 7\,x_{27} - 4\,x_{27}^{2} + x_{30} + x_{32} + 4\,x_{35} + 4\,x_{35}^{2} \\
& - x_{36} + x_{48} + 4\,x_{57} + 4\,x_{57}^{2} - x_{58} - 7\,x_{78} + x_{83} \\
& - 2\,x_{214}^{2}
\end{aligned}
$$

$$
Q_{17} = -1 + 2\,x_{10} + 10\,x_{10}^{2} - 7\,x_{10}\,x_{11} - 9\,x_{10}\,x_{12} - 8\,x_{10}\,x_{25}
$$

$$
\begin{aligned}
&+ 8\,x_{10}\,x_{27} + 24\,x_{10}\,x_{41} - x_{11} + 6\,x_{11}^2 + x_{11}\,x_{12} \\
&- 4\,x_{11}\,x_{27} - 8\,x_{12} - 3\,x_{12}^2 + 2\,x_{12}\,x_{13} - 8\,x_{12}\,x_{27} \\
&+ 16\,x_{12}\,x_{41} + x_{13}^2 + 4\,x_{21} - 3\,x_{25} + 2\,x_{26} - 4\,x_{27} \\
&+ 8\,x_{27}\,x_{41} + x_{33} + 4\,x_{35} + 4\,x_{35}^2 - x_{36} - 4\,x_{41} \\
&+ 4\,x_{41}^2 - x_{42} - x_{43} - 2\,x_{48} - 9\,x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{18} = 1 &+ x_1 - x_8\,x_9 - 4\,x_{32}^2 - 8\,x_{32}\,x_{33} - 4\,x_{33}^2 \\
&+ x_{34}
\end{aligned}
$$

$$
\begin{aligned}
Q_{19} = 1 &- 2\,x_{10} + 4\,x_{10}^2 + 4\,x_{10}\,x_{11} - 4\,x_{10}\,x_{12} - 8\,x_{10}\,x_{27} \\
&- 8\,x_{10}\,x_{65} - 20\,x_{11} - 7\,x_{11}^2 + 14\,x_{11}\,x_{12} + 8\,x_{11}\,x_{21} \\
&+ 16\,x_{11}\,x_{25} - 8\,x_{11}\,x_{27} - 16\,x_{11}\,x_{65} + x_{12} - 6\,x_{12}^2 \\
&+ 2\,x_{12}\,x_{13} + x_{13}^2 + 7\,x_{21} - 8\,x_{21}\,x_{25} + 8\,x_{21}\,x_{65} \\
&+ 2\,x_{25} + 8\,x_{25}\,x_{27} - 4\,x_{27} + 4\,x_{27}^2 - x_{30} + x_{35} - x_{43} \\
&- 4\,x_{65} + 4\,x_{65}^2 - x_{66} + 6\,x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{20} = &- 1 - x_1 + x_8\,x_9 - 5\,x_{10} - x_{10}^2 - 2\,x_{10}\,x_{12} + x_{11} \\
&+ x_{11}^2 + 2\,x_{11}\,x_{12} - 2\,x_{12} - x_{25} - x_{27} - 4\,x_{35} \\
&- 4\,x_{35}^2 + x_{36}
\end{aligned}
$$

$$
\begin{aligned}
Q_{21} = 1 &+ x_0 + x_1 - x_8\,x_9 + 11\,x_{10} + 12\,x_{10}^2 \\
&+ 9\,x_{10}\,x_{11} - 7\,x_{10}\,x_{12} - 8\,x_{10}\,x_{21} + 16\,x_{10}\,x_{22} \\
&+ 12\,x_{10}\,x_{25} + 8\,x_{10}\,x_{27} + 8\,x_{10}\,x_{30} + 6\,x_{11} - 13\,x_{11}^2 \\
&+ 5\,x_{11}\,x_{12} - 8\,x_{11}\,x_{21} + 64\,x_{11}\,x_{22} + 16\,x_{11}\,x_{25} \\
&+ 24\,x_{11}\,x_{27} + 16\,x_{11}\,x_{30} - 15\,x_{12} + 4\,x_{12}^2 - 2\,x_{12}\,x_{13} \\
&+ 4\,x_{12}\,x_{21} + 8\,x_{12}\,x_{27} - 4\,x_{12}\,x_{30} - x_{13}^2 + 17\,x_{21} \\
&- 20\,x_{21}^2 + 32\,x_{21}\,x_{22} - 8\,x_{21}\,x_{25} + 16\,x_{21}\,x_{27} \\
&+ 8\,x_{21}\,x_{30} + 2\,x_{22} - 16\,x_{22}^2 + 3\,x_{23} - 2\,x_{24} - 6\,x_{25} \\
&+ 4\,x_{25}^2 - 7\,x_{26} + 3\,x_{27} - 2\,x_{29} - 5\,x_{30} - 3\,x_{31} - 4\,x_{32} \\
&- 2\,x_{34} - 3\,x_{35} - 4\,x_{35}^2 - x_{36} + x_{37} - 4\,x_{38} - 2\,x_{39} \\
&+ x_{40} - 2\,x_{42} + x_{43} + x_{45} - 2\,x_{46} + x_{47} + x_{48} \\
&- x_{65} - 4\,x_{67} - 2\,x_{68} - 3\,x_{78} - x_{210}\,x_{223} - x_{214}^2 + x_{218}
\end{aligned}
$$

$$
\begin{aligned}
Q_{22} = 1 &+ x_0 + x_1 - x_8\,x_9 - 25\,x_{10}^2 + 8\,x_{10}\,x_{11} \\
&+ 8\,x_{10}\,x_{12} + 4\,x_{10}\,x_{25} - 24\,x_{10}\,x_{37} + 3\,x_{11}^2 - 2\,x_{11}\,x_{12} \\
&- x_{12} + x_{12}^2 - 5\,x_{21} - 2\,x_{25} - x_{26} - 3\,x_{27} + 4\,x_{35} \\
&+ 4\,x_{35}^2 - x_{36} + 2\,x_{37} - 4\,x_{37}^2 + x_{38} + x_{48} + 2\,x_{78} \\
&- x_{214}^2
\end{aligned}
$$

$$
Q_{23} = - x_{11}^2 - 4\,x_{11}\,x_{38} - 4\,x_{38}^2 + x_{39}
$$

$$
\begin{aligned}
Q_{24} = &- 2 - x_1 + x_2\,x_7 + x_3\,x_{10} + x_3\,x_{11} + x_4 \\
&- x_5\,x_{20} - 2\,x_{10} - 2\,x_{10}^2 - 4\,x_{10}\,x_{11} - 3\,x_{11} - x_{11}^2 \\
&+ 2\,x_{11}\,x_{12} + x_{12} + x_{12}^2 + x_{18} - 12\,x_{21} - 16\,x_{21}^2 \\
&- x_{25} + x_{40} - x_{83} - x_{84}
\end{aligned}
$$

$$
\begin{aligned}
Q_{25} = 1 &+ x_0 + x_{10} - x_{10}^2 + 5\,x_{10}\,x_{11} + 5\,x_{10}\,x_{12} \\
&+ 8\,x_{10}\,x_{27} + 18\,x_{11} - 13\,x_{11}^2 - 7\,x_{11}\,x_{12} - 16\,x_{11}\,x_{25}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\,x_{11}\,x_{27} - 24\,x_{11}\,x_{40} + 7\,x_{12} + x_{12}^2 - 8\,x_{12}\,x_{25} \\
&- 16\,x_{12}\,x_{40} - 6\,x_{21} + 5\,x_{25} - 12\,x_{25}^2 - 8\,x_{25}\,x_{27} \\
&- 8\,x_{25}\,x_{40} + 2\,x_{26} + 7\,x_{27} - 4\,x_{27}^2 + x_{30} - 4\,x_{35} \\
&- 4\,x_{35}^2 + x_{36} + 4\,x_{40} - 4\,x_{40}^2 + x_{41} - x_{48} - x_{78} \\
&- x_{214}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{26} = \; &- 1 + x_0 - x_1 + x_8\,x_9 - 3\,x_{10} - 5\,x_{10}^2 - 3\,x_{10}\,x_{11} \\
&- 11\,x_{10}\,x_{12} - 4\,x_{10}\,x_{25} - 8\,x_{10}\,x_{27} + 4\,x_{10}\,x_{57} - 5\,x_{11} \\
&+ 10\,x_{11}^2 + 5\,x_{11}\,x_{12} + 8\,x_{11}\,x_{25} + 4\,x_{11}\,x_{27} - 13\,x_{12} \\
&+ x_{12}^2 - 8\,x_{12}\,x_{25} - 24\,x_{12}\,x_{27} + 8\,x_{12}\,x_{57} - 4\,x_{21} \\
&+ x_{25} - 8\,x_{25}\,x_{27} + 3\,x_{26} - 9\,x_{27} - 8\,x_{27}^2 + x_{30} + x_{33} \\
&+ 8\,x_{35} + 8\,x_{35}^2 - 2\,x_{36} - 4\,x_{48} + 4\,x_{57} + 4\,x_{57}^2 - x_{58} \\
&- 5\,x_{78} - x_{214}^2
\end{aligned}
$$

$$
Q_{27} = \; - x_{12}^2 - 2\,x_{12}\,x_{13} - x_{13}^2 + x_{43}
$$

$$
\begin{aligned}
Q_{28} = 1 &+ x_1 - x_8\,x_9 + x_{10}^2 - 4\,x_{10}\,x_{43} + 2\,x_{12} \\
&- x_{12}^2 + 2\,x_{12}\,x_{13} - 8\,x_{12}\,x_{43} + x_{13}^2 - 2\,x_{27} + 3\,x_{43} \\
&- 4\,x_{43}^2 + x_{44}
\end{aligned}
$$

$$
\begin{aligned}
Q_{29} = \; &- 1 - x_1 + x_8\,x_9 - 4\,x_{10} + 5\,x_{10}^2 + 2\,x_{10}\,x_{11} \\
&- 6\,x_{10}\,x_{12} - 8\,x_{10}\,x_{27} - 4\,x_{10}\,x_{43} + 5\,x_{11} + x_{11}^2 \\
&- 8\,x_{11}\,x_{25} - 8\,x_{11}\,x_{27} - 13\,x_{12} + 3\,x_{12}^2 + 16\,x_{12}\,x_{27} \\
&- 8\,x_{12}\,x_{43} + 6\,x_{21} - 11\,x_{25} + 8\,x_{25}^2 - 2\,x_{26} + 12\,x_{27} \\
&- x_{30} + 4\,x_{35} + 4\,x_{35}^2 - x_{36} + 4\,x_{43} - 4\,x_{43}^2 + x_{44} \\
&+ x_{45} - 6\,x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{30} = \; &x_{10}\,x_{11} - x_{10}\,x_{12} - x_{11} - x_{11}\,x_{12} - 3\,x_{12} \\
&- 2\,x_{12}\,x_{13} - x_{13}^2 - 16\,x_{27}^2 - 32\,x_{27}\,x_{43} - 16\,x_{27}\,x_{44} \\
&- 16\,x_{27}\,x_{45} - 3\,x_{43} - 16\,x_{43}^2 - 16\,x_{43}\,x_{44} - 16\,x_{43}\,x_{45} \\
&- 2\,x_{44} - 4\,x_{44}^2 - 8\,x_{44}\,x_{45} - 4\,x_{45}^2 + x_{46} - x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{31} = 2 &+ x_1 - x_3\,x_{11} - x_4 + x_5\,x_{20} - x_8\,x_9 \\
&+ x_{10} - 12\,x_{10}^2 + 6\,x_{10}\,x_{11} + 4\,x_{10}\,x_{12} - 4\,x_{10}\,x_{25} \\
&+ 24\,x_{10}\,x_{27} + 16\,x_{10}\,x_{30} + 4\,x_{11} - x_{11}^2 - 8\,x_{11}\,x_{12} \\
&- 8\,x_{11}\,x_{25} - 8\,x_{11}\,x_{27} + x_{12} - 4\,x_{12}^2 + 2\,x_{12}\,x_{13} \\
&- 8\,x_{12}\,x_{27} - 4\,x_{12}\,x_{30} + x_{13}^2 - x_{18} + x_{21} - 8\,x_{25} \\
&+ 4\,x_{25}^2 + 8\,x_{25}\,x_{27} - x_{26} - 8\,x_{27} - 12\,x_{27}^2 - 16\,x_{27}\,x_{30} \\
&- x_{30} - 4\,x_{30}^2 - x_{43} + x_{47} + x_{48} + x_{83}
\end{aligned}
$$

$$
\begin{aligned}
Q_{32} = 1 &+ x_1 - x_8\,x_9 - x_{10}^2 - 2\,x_{10}\,x_{12} + x_{11}^2 \\
&+ 2\,x_{11}\,x_{12} - 4\,x_{12} + 8\,x_{12}\,x_{25} + x_{21} + 6\,x_{25} - 4\,x_{25}^2 \\
&+ x_{48} - 2\,x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{33} = 1 &- x_0 + x_3 - x_3\,x_{10} - x_3\,x_{11} + x_{10} \\
&+ x_{10}\,x_{11} + 5\,x_{10}\,x_{12} - 4\,x_{10}\,x_{25} - x_{11} - x_{11}^2 \\
&- 3\,x_{11}\,x_{12} + 5\,x_{12} - 3\,x_{12}^2 + 2\,x_{12}\,x_{13} + 8\,x_{12}\,x_{25} \\
&+ 12\,x_{12}\,x_{48} + x_{13}^2 - x_{21} - 10\,x_{25} - 8\,x_{25}^2
\end{aligned}
$$

$$
\begin{aligned}
&- 16\,x_{25}\,x_{48} + x_{26} - 2\,x_{27} - 4\,x_{35} - 4\,x_{35}^2 + x_{36} \\
&- x_{43} - 3\,x_{48} - 4\,x_{48}^2 + x_{49} + x_{78} + x_{83} + x_{84} \\
&+ x_{214}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{34} = 1 &+ x_1 - x_3\,x_{11} - x_8\,x_9 + 2\,x_{11} - 2\,x_{11}\,x_{12} \\
&- 4\,x_{11}\,x_{49} + x_{12} - 3\,x_{12}^2 - 8\,x_{12}\,x_{49} - x_{25} - 4\,x_{49}^2 \\
&+ x_{50} + x_{83}
\end{aligned}
$$

$$
Q_{35} = \; - 1 - 4\,x_{50} - 4\,x_{50}^2 + x_{51}
$$

$$
\begin{aligned}
Q_{36} = 1 &- x_{12}^2 - 8\,x_{12}\,x_{50} - 4\,x_{12}\,x_{51} + 8\,x_{50} - 8\,x_{50}^2 \\
&- 16\,x_{50}\,x_{51} - 2\,x_{51} - 4\,x_{51}^2 + x_{52}
\end{aligned}
$$

$$
\begin{aligned}
Q_{37} = 2 &+ x_1 - x_3\,x_{11} - x_8\,x_9 + 2\,x_{11} - 2\,x_{11}\,x_{12} \\
&- 4\,x_{11}\,x_{52} + x_{12} - 3\,x_{12}^2 - 8\,x_{12}\,x_{52} - x_{25} - 4\,x_{52}^2 \\
&+ x_{53} + x_{83}
\end{aligned}
$$

$$
Q_{38} = \; - 4\,x_{47}^2 - 8\,x_{47}\,x_{53} - 4\,x_{53}^2 + x_{54}
$$

$$
\begin{aligned}
Q_{39} = \; &- 1 - x_1 + x_8\,x_9 - x_{10} + x_{10}^2 - x_{10}\,x_{11} \\
&- x_{10}\,x_{12} + 4\,x_{10}\,x_{25} - 8\,x_{10}\,x_{47} - 4\,x_{10}\,x_{54} - x_{11}^2 \\
&- x_{11}\,x_{12} + x_{12} - x_{12}^2 - 3\,x_{25} - x_{26} - x_{27} \\
&+ 24\,x_{47} - 12\,x_{47}^2 + 8\,x_{47}\,x_{53} - 16\,x_{47}\,x_{54} + x_{48} \\
&+ 4\,x_{53}^2 + 11\,x_{54} - 4\,x_{54}^2 + x_{55} + x_{78}
\end{aligned}
$$

$$
Q_{40} = \; - 1 - 4\,x_{55} - 4\,x_{55}^2 + x_{56}
$$

$$
\begin{aligned}
Q_{41} = 1 &+ x_1 - x_8\,x_9 - x_{12}^2 - 8\,x_{12}\,x_{55} - 4\,x_{12}\,x_{56} \\
&+ 8\,x_{55} - 8\,x_{55}^2 - 16\,x_{55}\,x_{56} - 2\,x_{56} - 4\,x_{56}^2 + x_{57}
\end{aligned}
$$

$$
\begin{aligned}
Q_{42} = 1 &+ x_1 - x_8\,x_9 - 2\,x_{10} - x_{10}^2 - x_{10}\,x_{11} \\
&- 3\,x_{10}\,x_{12} - 4\,x_{10}\,x_{57} + x_{11}\,x_{12} - 3\,x_{12} - x_{12}^2 \\
&- 8\,x_{12}\,x_{57} + x_{21} - x_{25} - x_{27} - 4\,x_{57} - 4\,x_{57}^2 + x_{58} \\
&- x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{43} = 2 &- x_0 + x_1 - x_2\,x_7 - x_3\,x_{10} - x_3\,x_{11} \\
&- x_4 + x_5\,x_{20} - x_{18} - 12\,x_{58} - 4\,x_{58}^2 + x_{59} \\
&+ x_{83} + x_{84} + x_{214}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{44} = \; &- x_{10}\,x_{11} + x_{10}\,x_{12} - x_{11}^2 - x_{11}\,x_{12} + 2\,x_{12} \\
&- 2\,x_{12}^2 + 16\,x_{12}\,x_{58} + 8\,x_{12}\,x_{59} - 3\,x_{25} - 16\,x_{25}\,x_{58} \\
&- 8\,x_{25}\,x_{59} - x_{48} + 8\,x_{58} - 16\,x_{58}^2 - 16\,x_{58}\,x_{59} + 4\,x_{59} \\
&- 4\,x_{59}^2 + x_{60} + x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{45} = \; &- 1 - x_1 + x_8\,x_9 - x_{10} + 2\,x_{10}^2 + 2\,x_{10}\,x_{11} \\
&+ 2\,x_{10}\,x_{12} - 2\,x_{11} - 3\,x_{11}^2 + 2\,x_{11}\,x_{12} - 8\,x_{11}\,x_{25} \\
&- 4\,x_{11}\,x_{60} + 4\,x_{12} - 4\,x_{12}^2 + 2\,x_{12}\,x_{13} - 12\,x_{12}\,x_{48}
\end{aligned}
$$

$$
\begin{aligned}
&+ 16\,x_{12}\,x_{60} + x_{13}^2 - x_{21} - 5\,x_{25} + 4\,x_{25}^2 + 16\,x_{25}\,x_{48} \\
&- 16\,x_{25}\,x_{60} - x_{27} - x_{43} - x_{48} + 4\,x_{48}^2 - x_{49} \\
&- 4\,x_{60}^2 + x_{61}
\end{aligned}
$$

$$
\begin{aligned}
Q_{46} = 2 &- x_0 + x_1 - x_2\,x_7 - x_3\,x_{10} - x_3\,x_{11} \\
&- x_4 + x_5\,x_{20} - x_{18} - 12\,x_{53} - 4\,x_{53}^2 + x_{62} \\
&+ x_{83} + x_{84} + x_{214}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{47} = 1 &+ 4\,x_{47}^2 + 8\,x_{47}\,x_{53} - 12\,x_{53}^2 - 16\,x_{53}\,x_{61} \\
&- 16\,x_{53}\,x_{62} - x_{54} - 4\,x_{61}^2 - 8\,x_{61}\,x_{62} - 4\,x_{62}^2 + x_{63}
\end{aligned}
$$

$$
\begin{aligned}
Q_{48} = x_{12} &- x_{12}^2 - 8\,x_{12}\,x_{61} - 4\,x_{12}\,x_{63} - 16\,x_{61}^2 \\
&- 16\,x_{61}\,x_{63} - 4\,x_{63}^2 + x_{64}
\end{aligned}
$$

$$
\begin{aligned}
Q_{49} = 2 &+ x_1 - x_3\,x_{11} - x_8\,x_9 + 4\,x_{10} + 2\,x_{10}^2 \\
&+ 8\,x_{10}\,x_{21} + 3\,x_{11} - 11\,x_{11}^2 + 2\,x_{11}\,x_{12} + 16\,x_{11}\,x_{21} \\
&+ 2\,x_{12} + 2\,x_{12}^2 - 4\,x_{12}\,x_{21} - 10\,x_{21} - 4\,x_{21}^2 + x_{25} \\
&+ x_{65} - 4\,x_{78} + x_{83}
\end{aligned}
$$

$$
\begin{aligned}
Q_{50} = &\,2\,x_{10}\,x_{12} - 4\,x_{10}\,x_{25} + 8\,x_{10}\,x_{65} - 9\,x_{11} + 5\,x_{11}^2 \\
&+ 2\,x_{11}\,x_{12} + 16\,x_{11}\,x_{65} - 3\,x_{12} - x_{12}^2 + 4\,x_{12}\,x_{21} \\
&+ 6\,x_{21} - 8\,x_{21}\,x_{65} + x_{25} + x_{26} + x_{27} - x_{48} + 3\,x_{65} \\
&- 4\,x_{65}^2 + x_{66} + 4\,x_{78}
\end{aligned}
$$

$$
\begin{aligned}
Q_{51} = 1 &+ x_1 - x_8\,x_9 + 2\,x_{10} + x_{10}^2 - 3\,x_{10}\,x_{11} \\
&- 3\,x_{10}\,x_{12} + 8\,x_{10}\,x_{21} - 8\,x_{10}\,x_{25} + 16\,x_{10}\,x_{64} \\
&+ 16\,x_{10}\,x_{66} - 3\,x_{11} + x_{11}^2 - 3\,x_{11}\,x_{12} + 16\,x_{11}\,x_{21} \\
&+ 32\,x_{11}\,x_{64} + 32\,x_{11}\,x_{66} - 4\,x_{12} + 3\,x_{12}^2 - 2\,x_{12}\,x_{13} \\
&+ 12\,x_{12}\,x_{21} - x_{13}^2 - x_{21} - 4\,x_{21}^2 - 16\,x_{21}\,x_{64} \\
&- 16\,x_{21}\,x_{66} + 2\,x_{26} - 4\,x_{35} - 4\,x_{35}^2 + x_{36} + x_{43} \\
&- 2\,x_{48} - 4\,x_{64}^2 - 8\,x_{64}\,x_{66} - 3\,x_{65} - 2\,x_{66} - 4\,x_{66}^2 \\
&+ x_{67} + 5\,x_{78}
\end{aligned}
$$

$$
Q_{52} = -1 - x_1 + x_8\,x_9 - 4\,x_{67} - 4\,x_{67}^2 + x_{68}
$$

$$
Q_{53} = -x_3\,x_{15} + x_{69}
$$

$$
Q_{54} = x_{14} - x_{15} - 2\,x_{24}\,x_{69} - x_{70}
$$

$$
Q_{55} = x_{14} - x_{15} - 4\,x_{26}\,x_{69} - 2\,x_{29}\,x_{69} - x_{71}
$$

$$
\begin{aligned}
Q_{56} = &-x_3\,x_{15} - 2\,x_{31}\,x_{69} + 2\,x_{46}\,x_{69} - x_{69} - x_{72} \\
&+ x_{77}
\end{aligned}
$$

$$
Q_{57} = x_{14} - x_{15} - 4\,x_{32}\,x_{69} - 2\,x_{34}\,x_{69} - x_{73}
$$

$$
\begin{aligned}
Q_{58} = x_{14} &- x_{15} + 2\,x_{24}\,x_{69} - 2\,x_{31}\,x_{69} - 2\,x_{36}\,x_{69} \\
&+ x_{70} - x_{72} - x_{74}
\end{aligned}
$$

$$Q_{59} = x_{14} - x_{15} - 4\,x_{38}\,x_{69} - 2\,x_{39}\,x_{69} - x_{75}$$

$$Q_{60} = x_{14} - x_{15} - 2\,x_{42}\,x_{69} - x_{76}$$

$$Q_{61} = -2\,x_{3}\,x_{15} + x_{14} - x_{15} - 2\,x_{46}\,x_{69} - 2\,x_{69} - x_{77}$$

$$\begin{aligned} Q_{62} = {} & -x_{10}\,x_{11} + x_{10}\,x_{12} + x_{11} + x_{11}\,x_{12} - x_{12} \\ & - x_{12}^{2} + x_{78} \end{aligned}$$

$$Q_{63} = 1 + 2\,x_{13} + x_{13}^{2} - x_{78}^{2} + x_{79}$$

$$Q_{64} = -x_{79}^{2} + x_{80}$$

$$Q_{65} = x_{3}\,x_{15} - 4\,x_{67}\,x_{69} - 2\,x_{68}\,x_{69} + x_{69} + x_{81}$$

$$\begin{aligned} Q_{66} = {} & -x_{14}^{2} + 2\,x_{14}\,x_{15} + 2\,x_{14}\,x_{81} - x_{15}^{2} - 2\,x_{15}\,x_{81} \\ & - x_{81}^{2} + x_{82} \end{aligned}$$

$$Q_{67} = -1 - x_{3}\,x_{11} + x_{83}$$

$$Q_{68} = -x_{15}\,x_{83} + x_{85}\,x_{86} + x_{87}$$

$$Q_{69} = -1 + x_{3} - x_{3}\,x_{10} + x_{84}$$

$$\begin{aligned} Q_{70} = {} & -2 - x_{2}\,x_{7} + x_{3} + 2\,x_{3}\,x_{10} + 3\,x_{3}\,x_{11} \\ & - 2\,x_{3}\,x_{21} + x_{8}\,x_{9} + x_{83} + x_{85} \end{aligned}$$

$$Q_{71} = -x_{14}\,x_{84} + x_{86}$$

$$\begin{aligned} Q_{72} = {} & 3\,x_{14} + x_{14}\,x_{83} + 9\,x_{15} - 5\,x_{15}\,x_{83} - 4\,x_{15}\,x_{84} \\ & - x_{15}\,x_{85} - 2\,x_{24}\,x_{69} - 2\,x_{36}\,x_{69} - 2\,x_{42}\,x_{69} - x_{70} \\ & - x_{74} - x_{76} - 3\,x_{85}\,x_{86} - 3\,x_{87} + x_{88} \end{aligned}$$

$$Q_{73} = -x_{14}\,x_{70} + x_{15}\,x_{70} + x_{89}$$

$$Q_{74} = -x_{71}\,x_{89} + x_{90}$$

$$Q_{75} = -x_{72}\,x_{90} + x_{91}$$

$$Q_{76} = -x_{73}\,x_{91} + x_{92}$$

$$Q_{77} = -x_{74}\,x_{92} + x_{93}$$

$$Q_{78} = -x_{75}\,x_{93} + x_{94}$$

$$Q_{79} = -x_{76}\,x_{94} + x_{95}$$

$$Q_{80} = -x_{77}\,x_{95} + x_{96}$$

$$Q_{81} = -x_{87}\,x_{96} + x_{97}$$

$$Q_{82} = -x_{88}\,x_{97} + x_{98}$$

$$Q_{83} = -x_{98}^2 + x_{99}$$

$$Q_{84} = -x_2 - x_2\,x_{18} + x_{18} + x_{100}$$

$$Q_{85} = -x_{16}\,x_{100} + x_{101}$$

$$\begin{aligned} Q_{86} = {} & -2\,x_1^2 + 2\,x_1\,x_{14} + 2\,x_1\,x_{15} + 2\,x_1\,x_{101} + 2\,x_1\,x_{102} \\ & - x_{14}^2 - 2\,x_{14}\,x_{101} - x_{15}^2 - 2\,x_{15}\,x_{102} + x_{19}\,x_{103} \\ & - x_{80}\,x_{99} - x_{82}\,x_{99} - x_{101}^2 - x_{102}^2 \end{aligned}$$

$$Q_{87} = -x_2\,x_{17} + x_{17} - x_{17}\,x_{100} + x_{102}$$

$$Q_{88} = -x_{106} - 2\,x_{106}\,x_{237} + x_{108}$$

$$Q_{89} = -x_5\,x_{130} + x_{104}\,x_{228} - x_{107} - x_{108}\,x_{130}$$

$$Q_{90} = x_{109}\,x_{229} - x_{111}\,x_{133} - x_{112}$$

$$Q_{91} = x_{113}\,x_{229} - x_{114}\,x_{133} - x_{115}$$

$$Q_{92} = x_{116}\,x_{229} - x_{117}\,x_{133} - x_{118}$$

$$Q_{93} = x_{119}\,x_{229} - x_{120}\,x_{133} - x_{121}$$

$$Q_{94} = x_{122}\,x_{230} - x_{124}\,x_{133} - x_{125}$$

$$Q_{95} = x_{126}\,x_{230} - x_{127}\,x_{133} - x_{128}$$

$$Q_{96} = -x_{19}\,x_{130} + x_{133}$$

$$Q_{97} = x_{129} - x_{131}\,x_{133} - x_{132}$$

$$Q_{98} = -x_{133}\,x_{135} + x_{134} - x_{136}$$

$$Q_{99} = -x_{133}\,x_{138} + x_{137}\,x_{228} - x_{139}$$

$$Q_{100} = -x_{133}\,x_{141} + x_{140}\,x_{229} - x_{142}$$

$$Q_{101} = x_{143} - 2\,x_{143}\,x_{145} + x_{146}$$

$$Q_{102} = -x_{143}\,x_{146} - x_{146} + x_{147}$$

$$Q_{103} = x_{146}^2 - x_{147}^2 + x_{167}$$

$$Q_{104} = -x_{144}\,x_{149} - x_{149} + x_{150}$$

$$Q_{105} = 2\,x_{148} + x_{148}^2 + x_{149}^2 - x_{150}^2$$

$$Q_{106} = x_4^2 - 2\,x_4\,x_{144} + x_{144}^2 + x_{151} - x_{167}$$

$$Q_{107} = -1 - x_4\,x_{149} + x_6 + x_{144}\,x_{149} - x_{148} + x_{151}\,x_{152}$$

$$Q_{108} = 1 - 2\,x_5 + x_5^2 - 2\,x_5\,x_{144} + x_{153}$$

$$Q_{109} = -x_5\,x_{198} + x_{19} + x_{153}\,x_{154} - x_{197} + x_{199}$$

$$Q_{110} = -1 + 2\,x_{144}\,x_{213} - x_{155} + 2\,x_{213} - x_{213}^2$$

$$Q_{111} = 1 + x_{155}\,x_{156} - x_{197} - x_{198}\,x_{213} + x_{199} + 2\,x_{237}$$

$$Q_{112} = -x_{144}\,x_{158} - x_{158} + x_{159}$$

$$Q_{113} = 2\,x_{157} + x_{157}^2 + x_{158}^2 - x_{159}^2$$

$$\begin{aligned} Q_{114} = {} & -x_{144} + x_{144}^2 - 4\,x_{144}\,x_{237} - x_{155}\,x_{156} + x_{160} - x_{167} \\ & + x_{197} + x_{198}\,x_{213} - x_{199} + x_{215}\,x_{221} - x_{219} + 2\,x_{237} \\ & + 4\,x_{237}^2 \end{aligned}$$

$$\begin{aligned} Q_{115} = {} & -1 + x_{104} + x_{144}\,x_{158} - x_{157} - x_{158} - 2\,x_{158}\,x_{237} \\ & + x_{160}\,x_{161} \end{aligned}$$

$$\begin{aligned} Q_{116} = {} & -1 + x_{105} + x_{144}\,x_{149} - 2\,x_{144}\,x_{162} - x_{148} - 2\,x_{149}\,x_{237} \\ & + x_{160}\,x_{162} + x_{162} + 4\,x_{162}\,x_{237} \end{aligned}$$

$$Q_{117} = 1 + x_{144} - x_{144}\,x_{164} - x_{164} + x_{165}$$

$$\begin{aligned} Q_{118} = {} & -x_{144} + x_{144}\,x_{164} + 2\,x_{163} + x_{163}^2 - x_{164} + x_{164}^2 \\ & - x_{165} - x_{165}^2 \end{aligned}$$

$$\begin{aligned} Q_{119} = {} & x_{109} - x_{144} + x_{144}\,x_{164} + x_{160}\,x_{166} - x_{163} - x_{164} \\ & - 2\,x_{164}\,x_{237} + 2\,x_{237} \end{aligned}$$

$$Q_{120} = -x_{144} - x_{144}^2 + x_{167} + x_{215}\,x_{221} - x_{219}$$

$$\begin{aligned} Q_{121} = {} & x_{110} + x_{144}\,x_{150} - 2\,x_{144}\,x_{168} - x_{148} - 2\,x_{148}\,x_{237} \\ & - x_{149}\,x_{167} - 2\,x_{150}\,x_{237} + x_{155}\,x_{156} + x_{160}\,x_{168} + x_{168} \\ & + 4\,x_{168}\,x_{237} - x_{197} - x_{198}\,x_{213} + x_{199} \end{aligned}$$

$$Q_{122} = -x_{144}\,x_{170} - x_{170} + x_{171}$$

$$Q_{123} = 2\,x_{169} + x_{169}^2 + x_{170}^2 - x_{171}^2$$

$$\begin{aligned} Q_{124} = {} & -1 + x_{113} + x_{144}\,x_{170} + x_{160}\,x_{172} - x_{169} - x_{170} \\ & - 2\,x_{170}\,x_{237} \end{aligned}$$

$$Q_{125} = -x_{144}\,x_{174} - x_{174} + x_{175}$$

$$Q_{126} = 2\,x_{173} + x_{173}^2 + x_{174}^2 - x_{175}^2$$

$$\begin{aligned} Q_{127} = {} & -1 + x_{116} + x_{144}\,x_{174} + x_{160}\,x_{176} - x_{173} - x_{174} \\ & - 2\,x_{174}\,x_{237} \end{aligned}$$

$$Q_{128} = -x_{144}\,x_{178} - x_{178} + x_{179}$$

$$Q_{129} = 2\,x_{177} + x_{177}^2 + x_{178}^2 - x_{179}^2$$

$$\begin{aligned} Q_{130} = {} & -1 + x_{119} + x_{144}\,x_{178} + x_{160}\,x_{180} - x_{177} - x_{178} \\ & - 2\,x_{178}\,x_{237} \end{aligned}$$

$$Q_{131} = -x_{144}\,x_{182} - x_{182} + x_{183}$$

$$Q_{132} = 2\,x_{181} + x_{181}^2 + x_{182}^2 - x_{183}^2$$

$$\begin{aligned} Q_{133} = {} & -1 + x_{122} + x_{144}\,x_{182} + x_{160}\,x_{184} - x_{181} - x_{182} \\ & - 2\,x_{182}\,x_{237} \end{aligned}$$

$$Q_{134} = -x_{144}\,x_{186} - x_{186} + x_{187}$$

$$Q_{135} = 2\,x_{185} + x_{185}^2 + x_{186}^2 - x_{187}^2$$

$$\begin{aligned} Q_{136} = {} & -1 + x_{123} + x_{144}\,x_{186} - 2\,x_{144}\,x_{188} + x_{160}\,x_{188} - x_{185} \\ & - 2\,x_{186}\,x_{237} + x_{188} + 4\,x_{188}\,x_{237} \end{aligned}$$

$$Q_{137} = -x_{144}\,x_{190} - x_{190} + x_{191}$$

$$Q_{138} = 2\,x_{189} + x_{189}^2 + x_{190}^2 - x_{191}^2$$

$$\begin{aligned} Q_{139} = {} & -1 + x_{126} + x_{144}\,x_{190} + x_{160}\,x_{192} - x_{189} - x_{190} \\ & - 2\,x_{190}\,x_{237} \end{aligned}$$

$$Q_{140} = -x_{144}\,x_{194} - x_{194} + x_{195}$$

$$Q_{141} = 2\,x_{193} + x_{193}^2 + x_{194}^2 - x_{195}^2$$

$$\begin{aligned} Q_{142} = {} & -1 + x_{129} + x_{144}\,x_{194} + x_{160}\,x_{196} - x_{193} - x_{194} \\ & - 2\,x_{194}\,x_{237} \end{aligned}$$

$$Q_{143} = 1 - x_{144}\,x_{198} - x_{198} + x_{199}$$

$$\begin{aligned} Q_{144} = {} & -1 - x_5\,x_{198} + x_{19} + x_{153}\,x_{154} + x_{197} + x_{197}^2 \\ & + x_{198}^2 - x_{199} - x_{199}^2 \end{aligned}$$

$$\begin{aligned} Q_{145} = {} & x_{130} - 2\,x_{144}\,x_{200} + x_{160}\,x_{200} - x_{197} - x_{198} \\ & - 2\,x_{198}\,x_{237} + x_{199} + x_{200} + 4\,x_{200}\,x_{237} \end{aligned}$$

$$Q_{146} = -x_{144}\,x_{202} - x_{202} + x_{203}$$

$$Q_{147} = 2\,x_{201} + x_{201}^2 + x_{202}^2 - x_{203}^2$$

$$\begin{aligned} Q_{148} = {} & -1 + x_{134} + x_{144}\,x_{202} + x_{160}\,x_{204} - x_{201} - x_{202} \\ & - 2\,x_{202}\,x_{237} \end{aligned}$$

$$Q_{149} = -x_{144}\,x_{206} - x_{206} + x_{207}$$

$$Q_{150} = 2\,x_{205} + x_{205}^2 + x_{206}^2 - x_{207}^2$$

$$\begin{aligned} Q_{151} = {} & -1 + x_{137} + x_{144}\,x_{206} + x_{160}\,x_{208} - x_{205} - x_{206} \\ & - 2\,x_{206}\,x_{237} \end{aligned}$$

$$Q_{152} = -x_{144}\,x_{210} - x_{210} + x_{211}$$

$$Q_{153} = 2\,x_{209} + x_{209}^2 + x_{210}^2 - x_{211}^2$$

$$\begin{aligned} Q_{154} = {} & -1 + x_{140} + x_{144}\,x_{210} + x_{160}\,x_{212} - x_{209} - x_{210} \\ & - 2\,x_{210}\,x_{237} \end{aligned}$$

$$\begin{aligned} Q_{155} = {} & -1 - 2\,x_0 - 3\,x_0^2 - 2\,x_0\,x_4 - x_1 - x_1^2 - x_2 \\ & - x_2^2 - x_4 - 2\,x_4^2 - x_5^2 - x_5\,x_{20} + x_5\,x_{198} \\ & - x_6 + x_8\,x_9 + x_{10} + 2\,x_{10}\,x_{12} - x_{11}^2 - x_{12} \\ & - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 + x_{18} \\ & - x_{18}^2 - x_{19} - x_{19}^2 - x_{27} - x_{104}^2 - x_{105}^2 \\ & - x_{109}^2 - x_{110}^2 - x_{113}^2 - x_{116}^2 - x_{119}^2 - x_{122}^2 \\ & - x_{123}^2 - x_{126}^2 - x_{129}^2 - x_{130}^2 - x_{134}^2 - x_{137}^2 \\ & - x_{140}^2 + x_{143} - x_{144} + x_{144}^2 - 4\,x_{144}\,x_{237} + x_{149}^2 \\ & - x_{153}\,x_{154} + 2\,x_{155}\,x_{156} + x_{158}^2 + x_{160} - 2\,x_{164} + x_{164}^2 \\ & - x_{167} + x_{170}^2 + x_{174}^2 + x_{178}^2 + x_{182}^2 + x_{186}^2 \\ & + x_{190}^2 + x_{194}^2 - x_{197} + x_{198}^2 - 2\,x_{198}\,x_{213} + x_{199} \\ & + x_{202}^2 + x_{206}^2 - x_{210}^2 - x_{213}^2 + x_{215}\,x_{221} - x_{219} \\ & + x_{231}^2 + x_{232}^2 + x_{233} + x_{233}^2 - 4\,x_{237} - 4\,x_{237}^2 \end{aligned}$$

$$\begin{aligned} Q_{156} = {} & -x_5\,x_{10} - x_5\,x_{11} - x_5\,x_{12} - x_5\,x_{13} \\ & - x_5\,x_{14} - x_5\,x_{15} - x_5\,x_{16} - x_5\,x_{17} \\ & - x_5\,x_{18} + x_{213} \end{aligned}$$

$$Q_{157} = -2 + x_0 - 2\,x_1 - x_2 + 2\,x_2\,x_7 - x_3 - x_3\,x_{10}$$

$$
\begin{aligned}
&- x_3\,x_{11} + 2\,x_3\,x_{21} + x_4 - x_5\,x_{20} - x_6 \\
&+ x_6^2 + x_{11} + x_{11}^2 + 2\,x_{11}\,x_{12} + x_{12} + x_{12}^2 \\
&+ x_{18} - x_{25} - 3\,x_{83} - x_{84} - x_{85} - x_{144}\,x_{198} - x_{198} \\
&+ x_{199} - x_{214}^2 - x_{215}^2 - x_{216}^2 - x_{217}^2 - x_{218}^2 \\
&+ x_{224}^2 + x_{225}^2
\end{aligned}
$$

$$Q_{158} = \; - 1 + x_2 + x_6 - x_6^2$$

$$Q_{159} = \; - x_{144}\,x_{216} - x_{216} + x_{224}$$

$$Q_{160} = \; - 1 + x_{215}^2 + x_{216}^2 - x_{224}^2$$

$$Q_{161} = \; - x_{218} - x_{218}\,x_{219} + x_{225}$$

$$Q_{162} = \; - 1 + x_{217}^2 + x_{218}^2 - x_{225}^2$$

$$Q_{163} = \; - x_{210}^2 + x_{226}$$

$$Q_{164} = x_{216} - 2\,x_{220}\,x_{226} - x_{226}$$

$$Q_{165} = x_{144}\,x_{215} + x_{221}$$

$$Q_{166} = \; - x_{144} - x_{215}\,x_{221} + x_{219}$$

$$Q_{167} = \; - 1 - x_{209} - x_{215}\,x_{222} + x_{217}$$

$$
\begin{aligned}
Q_{168} = \; &- x_{10} - x_{11} - 4\,x_{21} - 2\,x_{24} - 4\,x_{26} - 2\,x_{29} - 2\,x_{31} \\
&- 4\,x_{32} - 2\,x_{34} - 2\,x_{36} - 4\,x_{38} - 2\,x_{39} - 2\,x_{42} - 2\,x_{46} \\
&- 4\,x_{67} - 2\,x_{68} - x_{210}\,x_{223} + x_{218}
\end{aligned}
$$

$$
\begin{aligned}
Q_{169} = \; &- 1 + 2\,x_0^2 + 2\,x_0\,x_4 - 2\,x_0\,x_{149} - 2\,x_0\,x_{158} + x_1^2 \\
&- 2\,x_1\,x_{198} + x_2^2 - 2\,x_2\,x_{186} + x_4^2 - 2\,x_4\,x_{158} \\
&+ x_{10}^2 - 2\,x_{10}\,x_{164} - 2\,x_{10}\,x_{210} + x_{11}^2 - 2\,x_{11}\,x_{170} \\
&- 2\,x_{11}\,x_{210} + x_{12}^2 - 2\,x_{12}\,x_{174} + x_{13}^2 - 2\,x_{13}\,x_{178} \\
&+ x_{14}^2 - 2\,x_{14}\,x_{182} + x_{15}^2 - 2\,x_{15}\,x_{190} + x_{16}^2 \\
&- 2\,x_{16}\,x_{194} + x_{17}^2 - 2\,x_{17}\,x_{202} + x_{18}^2 - 2\,x_{18}\,x_{206} \\
&- 8\,x_{21}\,x_{210} - 4\,x_{24}\,x_{210} - 8\,x_{26}\,x_{210} - 4\,x_{29}\,x_{210} \\
&- 4\,x_{31}\,x_{210} - 8\,x_{32}\,x_{210} - 4\,x_{34}\,x_{210} - 4\,x_{36}\,x_{210} \\
&- 8\,x_{38}\,x_{210} - 4\,x_{39}\,x_{210} - 4\,x_{42}\,x_{210} - 4\,x_{46}\,x_{210} \\
&- 8\,x_{67}\,x_{210} - 4\,x_{68}\,x_{210} + x_{144} - x_{144}\,x_{164} - x_{144}\,x_{227} \\
&+ x_{148}^2 + x_{157}^2 + x_{163}^2 + x_{164} + x_{165} + x_{169}^2 \\
&+ x_{173}^2 + x_{177}^2 + x_{181}^2 + x_{185}^2 + x_{189}^2 + x_{193}^2 \\
&+ x_{197}^2 + x_{201}^2 + x_{205}^2 + x_{209}^2 + x_{210} + x_{226} \\
&- x_{231}^2 - x_{232}^2 - x_{233} - x_{233}^2
\end{aligned}
$$

$$Q_{170} = \; - x_{105}\,x_{228} + x_{130}$$

$$Q_{171} = \; - x_{110}\,x_{229} + x_{130}$$

$$Q_{172} = -x_{123}\,x_{230} + x_{130}$$

$$\begin{aligned}
Q_{173} = {} & 2 + x_1 - x_2\,x_7 - x_3\,x_{10} - x_3\,x_{11} - x_4 \\
& + x_5\,x_{20} + x_{10} + x_{10}^2 + 2\,x_{10}\,x_{11} - 8\,x_{10}\,x_{21} \\
& - 4\,x_{10}\,x_{24} - 8\,x_{10}\,x_{26} - 4\,x_{10}\,x_{29} - 4\,x_{10}\,x_{31} \\
& - 8\,x_{10}\,x_{32} - 4\,x_{10}\,x_{34} - 4\,x_{10}\,x_{36} - 8\,x_{10}\,x_{38} \\
& - 4\,x_{10}\,x_{39} - 4\,x_{10}\,x_{42} - 4\,x_{10}\,x_{46} - 8\,x_{10}\,x_{67} \\
& - 4\,x_{10}\,x_{68} + 2\,x_{11} + 3\,x_{11}^2 - 2\,x_{11}\,x_{12} - 8\,x_{11}\,x_{21} \\
& - 4\,x_{11}\,x_{24} - 8\,x_{11}\,x_{26} - 4\,x_{11}\,x_{29} - 4\,x_{11}\,x_{31} \\
& - 8\,x_{11}\,x_{32} - 4\,x_{11}\,x_{34} - 4\,x_{11}\,x_{36} + 4\,x_{11}\,x_{38} \\
& - 4\,x_{11}\,x_{39} - 4\,x_{11}\,x_{42} - 4\,x_{11}\,x_{46} - 8\,x_{11}\,x_{67} \\
& - 4\,x_{11}\,x_{68} - x_{12} - x_{12}^2 - x_{18} + 8\,x_{21} - 16\,x_{21}\,x_{24} \\
& - 32\,x_{21}\,x_{26} - 16\,x_{21}\,x_{29} - 16\,x_{21}\,x_{31} - 32\,x_{21}\,x_{32} \\
& - 16\,x_{21}\,x_{34} - 16\,x_{21}\,x_{36} - 32\,x_{21}\,x_{38} - 16\,x_{21}\,x_{39} \\
& - 16\,x_{21}\,x_{42} - 16\,x_{21}\,x_{46} - 32\,x_{21}\,x_{67} - 16\,x_{21}\,x_{68} \\
& - 2\,x_{24} - 4\,x_{24}^2 - 16\,x_{24}\,x_{26} - 8\,x_{24}\,x_{29} - 8\,x_{24}\,x_{31} \\
& - 16\,x_{24}\,x_{32} - 8\,x_{24}\,x_{34} - 8\,x_{24}\,x_{36} - 16\,x_{24}\,x_{38} \\
& - 8\,x_{24}\,x_{39} - 8\,x_{24}\,x_{42} - 8\,x_{24}\,x_{46} - 16\,x_{24}\,x_{67} \\
& - 8\,x_{24}\,x_{68} + x_{25} - 4\,x_{26} - 12\,x_{26}^2 + 8\,x_{26}\,x_{28} \\
& - 16\,x_{26}\,x_{29} - 16\,x_{26}\,x_{31} - 32\,x_{26}\,x_{32} - 16\,x_{26}\,x_{34} \\
& - 16\,x_{26}\,x_{36} - 32\,x_{26}\,x_{38} - 16\,x_{26}\,x_{39} - 16\,x_{26}\,x_{42} \\
& - 16\,x_{26}\,x_{46} - 32\,x_{26}\,x_{67} - 16\,x_{26}\,x_{68} + 4\,x_{28}^2 - 3\,x_{29} \\
& - 4\,x_{29}^2 - 8\,x_{29}\,x_{31} - 16\,x_{29}\,x_{32} - 8\,x_{29}\,x_{34} - 8\,x_{29}\,x_{36} \\
& - 16\,x_{29}\,x_{38} - 8\,x_{29}\,x_{39} - 8\,x_{29}\,x_{42} - 8\,x_{29}\,x_{46} \\
& - 16\,x_{29}\,x_{67} - 8\,x_{29}\,x_{68} - 2\,x_{31} - 4\,x_{31}^2 - 16\,x_{31}\,x_{32} \\
& - 8\,x_{31}\,x_{34} - 8\,x_{31}\,x_{36} - 16\,x_{31}\,x_{38} - 8\,x_{31}\,x_{39} \\
& - 8\,x_{31}\,x_{42} - 8\,x_{31}\,x_{46} - 16\,x_{31}\,x_{67} - 8\,x_{31}\,x_{68} - 4\,x_{32} \\
& - 12\,x_{32}^2 + 8\,x_{32}\,x_{33} - 16\,x_{32}\,x_{34} - 16\,x_{32}\,x_{36} \\
& - 32\,x_{32}\,x_{38} - 16\,x_{32}\,x_{39} - 16\,x_{32}\,x_{42} - 16\,x_{32}\,x_{46} \\
& - 32\,x_{32}\,x_{67} - 16\,x_{32}\,x_{68} + 4\,x_{33}^2 - 3\,x_{34} - 4\,x_{34}^2 \\
& - 8\,x_{34}\,x_{36} - 16\,x_{34}\,x_{38} - 8\,x_{34}\,x_{39} - 8\,x_{34}\,x_{42} \\
& - 8\,x_{34}\,x_{46} - 16\,x_{34}\,x_{67} - 8\,x_{34}\,x_{68} - 2\,x_{36} - 4\,x_{36}^2 \\
& - 16\,x_{36}\,x_{38} - 8\,x_{36}\,x_{39} - 8\,x_{36}\,x_{42} - 8\,x_{36}\,x_{46} \\
& - 16\,x_{36}\,x_{67} - 8\,x_{36}\,x_{68} - 4\,x_{38} - 4\,x_{38}^2 - 16\,x_{38}\,x_{39} \\
& - 16\,x_{38}\,x_{42} - 16\,x_{38}\,x_{46} - 32\,x_{38}\,x_{67} - 16\,x_{38}\,x_{68} \\
& - 5\,x_{39} - 4\,x_{39}^2 - 8\,x_{39}\,x_{42} - 8\,x_{39}\,x_{46} - 16\,x_{39}\,x_{67} \\
& - 8\,x_{39}\,x_{68} - x_{40} - 2\,x_{42} - 4\,x_{42}^2 - 8\,x_{42}\,x_{46} \\
& - 16\,x_{42}\,x_{67} - 8\,x_{42}\,x_{68} - 2\,x_{46} - 4\,x_{46}^2 - 16\,x_{46}\,x_{67} \\
& - 8\,x_{46}\,x_{68} + 4\,x_{67} - 8\,x_{67}^2 - 16\,x_{67}\,x_{68} - 4\,x_{68} - 4\,x_{68}^2 \\
& + x_{83} + x_{84} - x_{149}^2 - x_{158}^2 + 2\,x_{164} - x_{164}^2 - x_{170}^2 \\
& - x_{174}^2 - x_{178}^2 - x_{182}^2 - x_{186}^2 - x_{190}^2 - x_{194}^2 \\
& - x_{198}^2 - x_{202}^2 - x_{206}^2 + x_{210}^2 - x_{210}\,x_{223} + x_{218} \\
& - x_{231}^2 - x_{232}^2 - x_{233} - x_{233}^2
\end{aligned}$$

$$\begin{aligned}
Q_{174} = {} & -1 - 4\,x_{107}^2 - 4\,x_{112}^2 - 4\,x_{115}^2 - 4\,x_{118}^2 - 4\,x_{121}^2 - 4\,x_{125}^2 \\
& - 4\,x_{128}^2 + x_{130}^2 - 4\,x_{132}^2 - 4\,x_{136}^2 - 4\,x_{139}^2 - 4\,x_{142}^2 \\
& - x_{234}^2 - x_{235}^2 - x_{236} - x_{236}^2 - 4\,x_{239}^2
\end{aligned}$$

$$Q_{175} = -x_5\,x_{238} + x_{130}$$

$$Q_{176} = x_3\,x_{130} - x_6\,x_{238} - x_{239}$$

B Coordinate names and artifact interpretation

The semantic names below describe the restored graph coordinates; the equation itself uses only the independent x_i. Recorded translation shifts must be applied before identifying an actual source quantity with a final coordinate. The names are annotations, not new equations.

Index	Source name	Index	Source name
0	t0	120	range3.quotient
1	x	121	normalized.range3.remainder
2	y	122	pow10.output
3	fact0.output	123	pow11.output
4	mul.1	124	range4.quotient
5	choose0.output	125	normalized.range4.remainder
6	pow0.output	126	pow12.output
7	endpoint.u0	127	range5.quotient
8	endpoint.ut	128	normalized.range5.remainder
9	mul.11	129	pow13.output
10	crt.encoded.a	130	pow14.output
11	crt.encoded.b	131	range6.quotient
12	crt.encoded.c	132	normalized.range6.remainder
13	crt.encoded.d	133	mul.430
14	crt.encoded.S	134	pow15.output
15	crt.encoded.Sp	135	range7.quotient
16	crt.encoded.u	136	normalized.range7.remainder
17	crt.encoded.vq	137	pow16.output
18	crt.encoded_index_minus_one	138	range8.quotient
19	pow1.output	139	normalized.range8.remainder
20	crt.index_quotient	140	pow17.output
21	mul.40	141	range9.quotient
22	add.50	142	normalized.range9.remainder
23	mul.53	143	shared.W
24	add.61	144	shared.A
25	mul.64	145	shared.Zminus1
26	add.71	146	mul.450
27	mul.75	147	mul.451
28	add.83	148	index0.X
29	mul.86	149	index0.Y
30	mul.96	150	mul.460
31	add.101	151	base0.T
32	add.112	152	pow0.quotient
33	add.119	153	base1.T
34	mul.123	154	pow1.quotient
35	mul.137	155	base2.T
36	add.142	156	pow2.quotient
37	mul.148	157	index2.X
38	add.156	158	index2.Y
39	mul.159	159	mul.503
40	add.172	160	base3.T
41	add.177	161	pow3.quotient
42	add.181	162	pow4.quotient
43	mul.184	163	index3.X
44	mul.188	164	index3.Y
45	mul.194	165	mul.529
46	add.199	166	pow5.quotient
47	add.205	167	index_addition.D
48	mul.210	168	pow6.quotient
49	mul.214	169	index5.X
50	add.221	170	index5.Y
51	mul.225	171	mul.554
52	mul.229	172	pow7.quotient
53	add.236	173	index6.X
54	mul.240	174	index6.Y
55	add.247	175	mul.567
56	mul.251	176	pow8.quotient
57	mul.255	177	index7.X

Index	Source name	Index	Source name
58	add.262	178	index7.Y
59	mul.266	179	mul.580
60	mul.270	180	pow9.quotient
61	add.277	181	index8.X
62	mul.281	182	index8.Y
63	mul.285	183	mul.593
64	add.292	184	pow10.quotient
65	add.299	185	index9.X
66	mul.303	186	index9.Y
67	add.308	187	mul.606
68	mul.311	188	pow11.quotient
69	mul.314	189	index10.X
70	sqd.1	190	index10.Y
71	sqd.2	191	mul.619
72	sqd.3	192	pow12.quotient
73	sqd.4	193	index11.X
74	sqd.5	194	index11.Y
75	sqd.6	195	mul.632
76	sqd.7	196	pow13.quotient
77	sqd.8	197	index12.X
78	sqd.9	198	index12.Y
79	mul.331	199	mul.645
80	sqd.11	200	pow14.quotient
81	mul.334	201	index13.X
82	sqd.12	202	index13.Y
83	mul.337	203	mul.658
84	mul.340	204	pow15.quotient
85	mul.342	205	index14.X
86	mul.344	206	index14.Y
87	sqd.13	207	mul.671
88	sqd.14	208	pow16.quotient
89	mul.350	209	index15.X
90	mul.351	210	index15.Y
91	mul.352	211	mul.684
92	mul.353	212	pow17.quotient
93	mul.354	213	self_radix.BH
94	mul.355	214	cofinal.horizon_root
95	mul.356	215	cap.H
96	mul.357	216	cap.J
97	mul.358	217	cap.S
98	mul.359	218	cap.R
99	mul.360	219	cap.B
100	mul.362	220	cap.e
101	mul.364	221	cap.f
102	mul.367	222	cap.g
103	crt.nonnegative_quotient	223	cap.h
104	pow3.output	224	cap.common_Ay
105	pow4.output	225	cap.BR
106	choose0.high	226	cap.Y2
107	normalized.choose0.remainder	227	cap.pooled_quotient
108	mul.396	228	remainder_budget.ratio0
109	pow5.output	229	remainder_budget.ratio1
110	pow6.output	230	remainder_budget.ratio2
111	range0.quotient	231	new_guard3.a
112	normalized.range0.remainder	232	new_guard3.b
113	pow7.output	233	new_guard3.c
114	range1.quotient	234	new_guard4.a
115	normalized.range1.remainder	235	new_guard4.b
116	pow8.output	236	new_guard4.c
117	range2.quotient	237	radix.odd_square_root_half
118	normalized.range2.remainder	238	bootstrap.normalizing_ratio
119	pow9.output	239	bootstrap.normalized_ceiling_error